

Vel Tech Group of Institutions

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2028_NeoColab_Competitive Coding II_L8

VTU_Week 9_MCQ

Attempt : 1

Total Mark : 15

Marks Obtained : 14

Section 1 : MCQ

1. In Backtracking, what is the base case for checking connectivity in an undirected graph?

Answer

All vertices are visited

Status : Correct

Marks : 1/1

2. Which of the following constraints are typically used in Sudoku backtracking?

Answer

All the mentioned options

Status : Correct

Marks : 1/1

3. Backtracking may lead to a solution that is _____.

Answer

suboptimal

Status : Correct

Marks : 1/1

4. In a Sudoku solver, when does the backtracking algorithm backtrack?

Answer

When no valid number can be placed in a cell

Status : Correct

Marks : 1/1

5. In the Hamiltonian Cycle problem using backtracking, what must be true for a path to be valid?

Answer

All vertices are included once, and the last connects to the first

Status : Correct

Marks : 1/1

6. Which of the following is essential for backtracking to function correctly?

Answer

A recursive call with a rollback (backtrack step)

Status : Correct

Marks : 1/1

7. What is the main idea behind Backtracking to check if a graph is connected?

Answer

Trying all possible paths from a starting vertex and marking visited vertices

Status : Correct

Marks : 1/1

8. What is the primary technique used to solve the Rat in a Maze problem?

Answer

Backtracking

Status : Correct

Marks : 1/1

9. In Backtracking, what is the purpose of marking vertices as visited during graph traversal?

Answer

To avoid revisiting the same vertex

Status : Correct

Marks : 1/1

10. In backtracking solutions for maze problems, when do we backtrack from a cell?

Answer

When all directions from the current cell are either visited or blocked

Status : Correct

Marks : 1/1

11. In general, backtracking can be used to solve?

Answer

Combinatorial problems

Status : Correct

Marks : 1/1

12. Which data structure is commonly used in backtracking to store partial solutions?

Answer

Stack or List

Status : Correct

Marks : 1/1

13. Which of the following best describes the backtracking approach to generating permutations of a string?

Answer

Fix one character, recursively permute the rest, and backtrack

Status : Correct

Marks : 1/1

14. Which of the following problems cannot be solved using backtracking?

Answer

Finding shortest path in weighted graph

Status : Wrong

Marks : 0/1

15. What is a base case in a backtracking algorithm?

Answer

A condition when a solution is found or invalid

Status : Correct

Marks : 1/1